

RAJ KALPESH MATHURIA

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🌐 <https://www.linkedin.com/in/raj-mathuria-98a710283/> 📜 Certifications

AI/ML Researcher with 1.5+ years in deep learning, computer vision, and scalable ML. Research experience at ISRO in spatio-temporal modeling and satellite-ground data fusion, with Elsevier, Springer, and IEEE publications involving NASA and ISRO. Experienced in PyTorch, TensorFlow, distributed training, and deploying ML systems.

Education

Sardar Patel Institute of Technology (SPIT)

B.Tech in Computer Engineering (Minor in Banking Technology)

August 2023 – August 2027

CGPA: 8.09 / Minor CGPA: 9.00

Indian Institute of Technology Madras (IIT Madras)

Bachelor of Science in Data Science and Applications

November 2023 – January 2027

CGPA: 8.67 / Project CGPA: 10.00

Technical Skills

Programming: Python, C, C++, Java, SQL

ML & AI: PyTorch, Scikit-learn, Deep Learning, Computer Vision, NLP, LLMs, RAG, Generative AI, Explainable AI

Scientific ML & Data: Spatio-temporal Modeling, Satellite Data, Data Fusion, Time-Series Forecasting, NumPy, Pandas

Systems & Tools: Distributed Training, DDP, Model Optimization, OpenCV, Hugging Face, LangChain, Docker, Azure

Experience

Barclays | Technology Summer Intern

Pune, Maharashtra

June 2026 – August 2026

Java, Spring Boot, Microservices

- Modernized backend workflows with configurable orchestration, reducing maintenance overhead by **40%**.
- Delivered **2 production REST APIs** and a full-stack log analyzer supporting **10+ debugging operations**.
- Merged **2 endpoints** into production, reducing integration effort and earning a Full-Time return Offer.

Systems Plus | GenAI Developer Intern

Remote

February 2026 – April 2026

Python, Azure OpenAI, LangChain

- Architected scalable AI systems for recruitment and document intelligence, increasing process efficiency by **35%**.
- Constructed **3+ AI systems** using Azure OpenAI, LangChain, RAG, and Python for enterprise automation.
- Deployed **5+ production-ready AI solutions**, reducing manual effort by **60%-80%** and securing a Full-Time offer.

Research & Development

ISRO - Space Applications Centre | Tech Research Lead

Ahmedabad, Gujarat

December 2024 – July 2025

Python, GRU, Spatio-temporal Fusion

- Innovated StEFFuRNeT (**Attention-GRU**) to fuse satellite and ground data for air quality forecasting ($R^2 = 0.990$).
- Engineered a **spatio-temporal fusion pipeline** with cyclic encoding to enable accurate **24-72-hour** forecast windows.
- Achieved a **MAE = 0.769 $\mu\text{g}/\text{m}^3$** for NO_2 prediction, improving accuracy through satellite-ground data integration.

India's Intelligence Agency HQ (NTRO) | Research Intern

New Delhi, Delhi

August 2025 – November 2025

Python, Flask, Electron, Forensics

- Decoded **2 proprietary Hikvision filesystem structures** to recover video metadata and reconstruct forensic timelines.
- Built a **Flask pipeline** supporting 3 disk formats with real-time H.264 to MP4 conversion and cached streaming.
- Packaged the forensic viewer for **3 desktop platforms** using Electron, enabling cross-platform DVR evidence analysis.

Projects & Achievements

Smart India Hackathon (SIH) Winner | Team Lead & GenAI Developer 🏆

September 2024 – January 2025

- Led a team of 6 to build a deepfake detection system using Computer Vision and AI/ML, achieving **98% accuracy**.
- Refined CNN training with **DDP**, cutting training time by **50%** achieving **98.6% accuracy** for audio anomaly detection.
- Boosted runtime efficiency by 30% and won **1st Place nationally among 87,000 teams** for real-time detection.

RadarCast-Net (ISRO-BAH Hackathon) | Team Lead, AI/ML & Backend Developer 🏆

June 2024 – August 2024

- Enhanced deep learning precipitation nowcasting using **optical flow and Kalman filtering**, achieving 9.18% error.
- Combined **CNN-LSTM** architectures to optimize weight distribution and designed a Python UI for seamless visualization.
- Achieved **90.82% prediction accuracy**, outperforming traditional models for real-time radar-based forecasting.

KisanVerse (Meta/The Nudge Top 10) | Team Lead & Fullstack Developer 🏆

March 2025 – July 2025

- Integrated an AI-driven agricultural platform to assist farmers with crop planning and intelligent market access.
- Deployed **ResNet50** for soil classification (94.34%) and **CNNs** for plant disease detection (99.44%) with RAG interfaces.
- Created a **voice-based interaction system** for government subsidy applications and crop recommendations.

Publications

- Elsevier Publication (APR): Downscaling Satellite Nitrogen Dioxide – with NASA & ISRO in Q1 journal.
- IEEE Publication: Explainable AI-Based Intrusion Detection for Cybersecurity– FMLDS 2026, Kobe, Japan.